1. Student Database-

```
bmscecse@bmscecse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~$ cqlsh
Connected to Test Cluster at 127.0.0.1:9042
[cqlsh 6.1.0 | Cassandra 4.1.8 | CQL spec 3.4.6 | Native protocol v5]
Use HELP for help.
cqlsh> CREATE KEYSPACE colleges
   ... WITH replication = {
   ...     'class': 'SimpleStrategy',
   ...     'replication_factor': 1
   ... };
cqlsh> USE colleges;
cqlsh:colleges> CREATE TABLE Student_Info (
            ...     USN text PRIMARY KEY,
            ...     Name text,
            ...     Branch text,
            ...     Semester int,
            ...     CGPA decimal
            ... );
cqlsh:colleges> INSERT INTO Student_Info (USN, Name, Branch, Semester, CGPA)
            ... VALUES ('1MS21CS001', 'Rahul', 'CSE', 5, 8.5);
cqlsh:colleges>
cqlsh:colleges> INSERT INTO Student_Info (USN, Name, Branch, Semester, CGPA)
            ... VALUES ('1MS21EC002', 'Sneha', 'ECE', 4, 8.8);
cqlsh:colleges>
cqlsh:colleges> INSERT INTO Student_Info (USN, Name, Branch, Semester, CGPA)
            ... VALUES ('1MS21ME003', 'Arjun', 'ME', 6, 7.9);
cqlsh:colleges> UPDATE Student_Info
            ... SET CGPA = 9.1
            ... WHERE USN = '1MS21CS001';
cqlsh:colleges> SELECT * FROM Student_Info;

 usn        | branch | cgpa | name  | semester
------------+--------+------+-------+----------
 1MS21ME003 |     ME |  7.9 | Arjun |        6
 1MS21EC002 |    ECE |  8.8 | Sneha |        4
 1MS21CS001 |    CSE |  9.1 | Rahul |        5

(3 rows)
cqlsh:colleges>
```

2. Book management system-

```
bmscecse@bmscecse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~$ cqlsh
Connected to Test Cluster at 127.0.0.1:9042
[cqlsh 6.1.0 | Cassandra 4.1.8 | CQL spec 3.4.6 | Native protocol v5]
Use HELP for help.
cqlsh> CREATE KEYSPACE libraryy
   ... WITH replication = {
   ...     'class': 'SimpleStrategy',
   ...     'replication_factor': 1
   ... };
cqlsh> USE libraryy;
cqlsh:libraryy> CREATE TABLE Book_Info (
            ...     Book_Id text PRIMARY KEY,
            ...     Title text,
            ...     Author text,
            ...     Price decimal,
            ...     Category text
            ... );
cqlsh:libraryy> BEGIN BATCH
            ...     INSERT INTO Book_Info (Book_Id, Title, Author, Price, Category)
            ...     VALUES ('B001', 'The Catcher in the Rye', 'J.D. Salinger', 15.99, 'Fiction');
            ...
            ...     INSERT INTO Book_Info (Book_Id, Title, Author, Price, Category)
            ...     VALUES ('B002', '1984', 'George Orwell', 18.50, 'Dystopian');
            ...
            ...     INSERT INTO Book_Info (Book_Id, Title, Author, Price, Category)
            ...     VALUES ('B003', 'To Kill a Mockingbird', 'Harper Lee', 12.99, 'Fiction');
            ...
            ...     INSERT INTO Book_Info (Book_Id, Title, Author, Price, Category)
            ...     VALUES ('B004', 'The Great Gatsby', 'F. Scott Fitzgerald', 14.99, 'Classic');
            ... APPLY BATCH;
cqlsh:libraryy> UPDATE Book_Info
            ... SET Price = 20.00
            ... WHERE Book_Id = 'B002';
cqlsh:libraryy> DELETE FROM Book_Info
            ... WHERE Book_Id = 'B003';
cqlsh:libraryy> SELECT * FROM Book_Info;

 book_id | author              | category  | price | title
---------+---------------------+-----------+-------+-----------------------
    B002 |       George Orwell | Dystopian | 20.00 |                  1984
    B004 | F. Scott Fitzgerald |   Classic | 14.99 |       The Great Gatsby
    B001 |       J.D. Salinger |   Fiction | 15.99 | The Catcher in the Rye

(3 rows)
cqlsh:libraryy>
```

3. Customer orders

```
bmscecse@bmscecse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~$ cqlsh
Connected to Test Cluster at 127.0.0.1:9042
[cqlsh 6.1.0 | Cassandra 4.1.8 | CQL spec 3.4.6 | Native protocol v5]
Use HELP for help.
cqlsh> CREATE KEYSPACE Ecommercee
   ... WITH replication = {
   ...     'class': 'SimpleStrategy',
   ...     'replication_factor': 1
   ... };
cqlsh> USE Ecommercee;
cqlsh:ecommercee> CREATE TABLE Customer_Orders (
              ...     Order_Id text PRIMARY KEY,
              ...     Customer_Name text,
              ...     Product_Name text,
              ...     Quantity int,
              ...     Order_Date date
              ... );
cqlsh:ecommercee> INSERT INTO Customer_Orders (Order_Id, Customer_Name, Product_Name, Quantity, Order_Date)
              ... VALUES ('O1001', 'John Doe', 'Laptop', 1, '2023-10-01');
cqlsh:ecommercee>
cqlsh:ecommercee> INSERT INTO Customer_Orders (Order_Id, Customer_Name, Product_Name, Quantity, Order_Date)
              ... VALUES ('O1002', 'Jane Smith', 'Smartphone', 2, '2023-10-03');
cqlsh:ecommercee>
cqlsh:ecommercee> INSERT INTO Customer_Orders (Order_Id, Customer_Name, Product_Name, Quantity, Order_Date)
              ... VALUES ('O1003', 'Alice Johnson', 'Headphones', 3, '2023-10-05');
cqlsh:ecommercee> ALTER TABLE Customer_Orders
              ... ADD Delivery_Status text;
cqlsh:ecommercee> UPDATE Customer_Orders
              ... SET Delivery_Status = 'Delivered'
              ... WHERE Order_Id = 'O1002';
cqlsh:ecommercee> SELECT * FROM Customer_Orders;

 order_id | customer_name | delivery_status | order_date | product_name | quantity
----------+---------------+-----------------+------------+--------------+----------
    O1002 |    Jane Smith |       Delivered | 2023-10-03 |   Smartphone |        2
    O1001 |      John Doe |            null | 2023-10-01 |       Laptop |        1
    O1003 | Alice Johnson |            null | 2023-10-05 |   Headphones |        3

(3 rows)
cqlsh:ecommercee>
```

4. Employee skills-

```
bmscecse@bmscecse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~$ cqlsh
Connected to Test Cluster at 127.0.0.1:9042
[cqlsh 6.1.0 | Cassandra 4.1.8 | CQL spec 3.4.6 | Native protocol v5]
Use HELP for help.
cqlsh> CREATE KEYSPACE Company
   ... WITH replication = {
   ...     'class': 'SimpleStrategy',
   ...     'replication_factor': 1
   ... };
cqlsh> USE Company;
cqlsh:company> CREATE TABLE Employee_Skills (
           ...     Emp_Id text PRIMARY KEY,
           ...     Emp_Name text,
           ...     Skills SET<text>
           ... );
cqlsh:company> INSERT INTO Employee_Skills (Emp_Id, Emp_Name, Skills)
           ... VALUES ('E001', 'John Doe', {'Java', 'C++', 'SQL'});
cqlsh:company>
cqlsh:company> INSERT INTO Employee_Skills (Emp_Id, Emp_Name, Skills)
           ... VALUES ('E002', 'Jane Smith', {'Python', 'Machine Learning', 'Data Analysis'});
cqlsh:company>
cqlsh:company> INSERT INTO Employee_Skills (Emp_Id, Emp_Name, Skills)
           ... VALUES ('E003', 'Alice Johnson', {'JavaScript', 'HTML', 'CSS', 'React'});
cqlsh:company> UPDATE Employee_Skills
           ... SET Skills = Skills + {'Docker'}
           ... WHERE Emp_Id = 'E002';
cqlsh:company> SELECT Emp_Name, Skills FROM Employee_Skills;

 emp_name      | skills
---------------+-------------------------------------------------------
      John Doe |                                  {'C++', 'Java', 'SQL'}
    Jane Smith | {'Data Analysis', 'Docker', 'Machine Learning', 'Python'}
 Alice Johnson |                    {'CSS', 'HTML', 'JavaScript', 'React'}

(3 rows)
cqlsh:company>
```

5. Employee skills-

```
bmscecse@bmscecse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~$ cqlsh
Connected to Test Cluster at 127.0.0.1:9042
[cqlsh 6.1.0 | Cassandra 4.1.8 | CQL spec 3.4.6 | Native protocol v5]
Use HELP for help.
cqlsh> CREATE KEYSPACE Store
   ... WITH replication = {
   ...     'class': 'SimpleStrategy',
   ...     'replication_factor': 1
   ... };
cqlsh> USE Store;
cqlsh:store> CREATE TABLE Product_Info (
         ...     Product_Id text PRIMARY KEY,
         ...     Product_Name text,
         ...     Price decimal,
         ...     Category text
         ... );
cqlsh:store> INSERT INTO Product_Info (Product_Id, Product_Name, Price, Category)
         ... VALUES ('P001', 'Smartphone', 299.99, 'Electronics')
         ... USING TTL 20;
cqlsh:store> SELECT * FROM Product_Info WHERE Product_Id = 'P001';

 product_id | category    | price  | product_name
------------+-------------+--------+--------------
       P001 | Electronics | 299.99 |   Smartphone

(1 rows)
cqlsh:store> SELECT * FROM Product_Info WHERE Product_Id = 'P001';

 product_id | category | price | product_name
------------+----------+-------+--------------

(0 rows)
cqlsh:store>
```

6. Online course registeration

```
bmscecse@bmscecse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~$ cqlsh
Connected to Test Cluster at 127.0.0.1:9042
[cqlsh 6.1.0 | Cassandra 4.1.8 | CQL spec 3.4.6 | Native protocol v5]
Use HELP for help.
cqlsh> CREATE KEYSPACE Online_Courses
   ... WITH replication = {
   ...     'class': 'SimpleStrategy',
   ...     'replication_factor': 1
   ... };
cqlsh> USE Online_Courses;
cqlsh:online_courses> CREATE TABLE Course_Registration (
                   ...     Reg_Id text PRIMARY KEY,
                   ...     Student_Name text,
                   ...     Course_Name text,
                   ...     Registration_Date date
                   ... );
cqlsh:online_courses> INSERT INTO Course_Registration (Reg_Id, Student_Name, Course_Name, Registration_Date)
                   ... VALUES ('R001', 'John Doe', 'Data Science', '2023-10-01');
cqlsh:online_courses>
cqlsh:online_courses> INSERT INTO Course_Registration (Reg_Id, Student_Name, Course_Name, Registration_Date)
                   ... VALUES ('R002', 'Jane Smith', 'Machine Learning', '2023-10-02');
cqlsh:online_courses>
cqlsh:online_courses> INSERT INTO Course_Registration (Reg_Id, Student_Name, Course_Name, Registration_Date)
                   ... VALUES ('R003', 'Alice Johnson', 'Web Development', '2023-10-03');
cqlsh:online_courses> ALTER TABLE Course_Registration
                   ... ADD Payment_Status text;
cqlsh:online_courses> UPDATE Course_Registration
                   ... SET Payment_Status = 'Paid'
                   ... WHERE Reg_Id = 'R001';
cqlsh:online_courses> INSERT INTO Course_Registration (Reg_Id, Student_Name, Course_Name, Registration_Date)
                   ... VALUES ('R004', 'Tom Hanks', 'Cloud Computing', '2023-10-04')
                   ... USING TTL 30;
cqlsh:online_courses> SELECT * FROM Course_Registration WHERE Reg_Id = 'R004';

 reg_id | course_name     | payment_status | registration_date | student_name
--------+-----------------+----------------+-------------------+--------------
   R004 | Cloud Computing |           null |        2023-10-04 |    Tom Hanks

(1 rows)
cqlsh:online_courses> SELECT * FROM Course_Registration WHERE Reg_Id = 'R004';

 reg_id | course_name | payment_status | registration_date | student_name
--------+-------------+----------------+-------------------+--------------

(0 rows)
cqlsh:online_courses>
```

7. Restaurant Orders

```
bmscecse@bmscecse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~$ cqlsh
Connected to Test Cluster at 127.0.0.1:9042
[cqlsh 6.1.0 | Cassandra 4.1.8 | CQL spec 3.4.6 | Native protocol v5]
Use HELP for help.
cqlsh> CREATE KEYSPACE Restaurant
   ... WITH replication = {
   ...     'class': 'SimpleStrategy',
   ...     'replication_factor': 1
   ... };
cqlsh> USE Restaurant;
cqlsh:restaurant> CREATE TABLE Orders (
              ...     Order_Id text PRIMARY KEY,
              ...     Customer_Name text,
              ...     Item_Name text,
              ...     Quantity int,
              ...     Bill_Amount decimal
              ... );
cqlsh:restaurant> BEGIN BATCH
              ...     INSERT INTO Orders (Order_Id, Customer_Name, Item_Name, Quantity, Bill_Amount)
              ...     VALUES ('O001', 'John Doe', 'Pizza', 2, 20.00);
              ...
              ...     INSERT INTO Orders (Order_Id, Customer_Name, Item_Name, Quantity, Bill_Amount)
              ...     VALUES ('O002', 'Jane Smith', 'Pasta', 1, 12.50);
              ...
              ...     INSERT INTO Orders (Order_Id, Customer_Name, Item_Name, Quantity, Bill_Amount)
              ...     VALUES ('O003', 'Alice Johnson', 'Burger', 3, 15.75);
              ... APPLY BATCH;
cqlsh:restaurant> UPDATE Orders
              ... SET Bill_Amount = 14.00
              ... WHERE Order_Id = 'O002';
cqlsh:restaurant> DELETE FROM Orders
              ... WHERE Order_Id = 'O003';
cqlsh:restaurant> SELECT * FROM Orders;

 order_id | bill_amount | customer_name | item_name | quantity
----------+-------------+---------------+-----------+----------
     O001 |       20.00 |      John Doe |     Pizza |        2
     O002 |       14.00 |    Jane Smith |     Pasta |        1

(2 rows)
cqlsh:restaurant>
```

8. Movie Ratings System

```
bmscecse@bmscecse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~$ cqlsh
Connected to Test Cluster at 127.0.0.1:9042
[cqlsh 6.1.0 | Cassandra 4.1.8 | CQL spec 3.4.6 | Native protocol v5]
Use HELP for help.
cqlsh> CREATE KEYSPACE Movies
   ... WITH replication = {
   ...     'class': 'SimpleStrategy',
   ...     'replication_factor': 1
   ... };
cqlsh> USE Movies;
cqlsh:movies> CREATE TABLE Movie_Ratings (
          ...     Movie_Id text PRIMARY KEY,
          ...     Movie_Name text,
          ...     Ratings MAP<text, int>
          ... );
cqlsh:movies> INSERT INTO Movie_Ratings (Movie_Id, Movie_Name, Ratings)
          ... VALUES ('M001', 'Inception', {'John': 5, 'Jane': 4, 'Alice': 5});
cqlsh:movies>
cqlsh:movies> INSERT INTO Movie_Ratings (Movie_Id, Movie_Name, Ratings)
          ... VALUES ('M002', 'The Matrix', {'John': 4, 'Bob': 5, 'Charlie': 4});
cqlsh:movies> UPDATE Movie_Ratings
          ... SET Ratings = Ratings + {'Tom': 4}
          ... WHERE Movie_Id = 'M001';
cqlsh:movies> UPDATE Movie_Ratings
          ... SET Ratings = Ratings + {'John': 3}
          ... WHERE Movie_Id = 'M001';
cqlsh:movies> SELECT Ratings FROM Movie_Ratings WHERE Movie_Id = 'M001';

 ratings
---------------------------------------------
 {'Alice': 5, 'Jane': 4, 'John': 3, 'Tom': 4}

(1 rows)
cqlsh:movies>
```

9. Shopping Cart System

```
bmscecse@bmscecse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~$ cqlsh
Connected to Test Cluster at 127.0.0.1:9042
[cqlsh 6.1.0 | Cassandra 4.1.8 | CQL spec 3.4.6 | Native protocol v5]
Use HELP for help.
cqlsh> CREATE KEYSPACE Shopping
   ... WITH replication = {
   ...     'class': 'SimpleStrategy',
   ...     'replication_factor': 1
   ... };
cqlsh> USE Shopping;
cqlsh:shopping> CREATE TABLE Cart (
            ...     Cart_Id text PRIMARY KEY,
            ...     User_Name text,
            ...     Items LIST<text>
            ... );
cqlsh:shopping> INSERT INTO Cart (Cart_Id, User_Name, Items)
            ... VALUES ('C001', 'JohnDoe', ['Laptop', 'Smartphone', 'Headphones']);
cqlsh:shopping> UPDATE Cart
            ... SET Items = Items + ['Tablet']
            ... WHERE Cart_Id = 'C001';
cqlsh:shopping> UPDATE Cart
            ... SET Items = Items - ['Smartphone']
            ... WHERE Cart_Id = 'C001';
cqlsh:shopping> INSERT INTO Cart (Cart_Id, User_Name, Items)
            ... VALUES ('C002', 'JaneSmith', ['TV', 'Fridge', 'Washing Machine'])
            ... USING TTL 60;
cqlsh:shopping> SELECT * FROM Cart WHERE Cart_Id = 'C002';

 cart_id | items                                | user_name
---------+--------------------------------------+-----------
    C002 | ['TV', 'Fridge', 'Washing Machine'] | JaneSmith

(1 rows)
cqlsh:shopping> SELECT * FROM Cart WHERE Cart_Id = 'C002';

 cart_id | items | user_name
---------+-------+-----------

(0 rows)
cqlsh:shopping>
```

10. Bank Account Balance

```
bmscecse@bmscecse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~$ cqlsh
Connected to Test Cluster at 127.0.0.1:9042
[cqlsh 6.1.0 | Cassandra 4.1.8 | CQL spec 3.4.6 | Native protocol v5]
Use HELP for help.
cqlsh> CREATE KEYSPACE Banking
   ... WITH replication = {
   ...     'class': 'SimpleStrategy',
   ...     'replication_factor': 1
   ... };
cqlsh> USE Banking;
cqlsh:banking> CREATE TABLE Account_Transactions (
           ...     Account_No text PRIMARY KEY,
           ...     Balance COUNTER
           ... );
cqlsh:banking> UPDATE Account_Transactions
           ... SET Balance = Balance + 100
           ... WHERE Account_No = 'A123';
cqlsh:banking>
cqlsh:banking> UPDATE Account_Transactions
           ... SET Balance = Balance + 200
           ... WHERE Account_No = 'A123';
cqlsh:banking> UPDATE Account_Transactions
           ... SET Balance = Balance - 150
           ... WHERE Account_No = 'A123';
cqlsh:banking> SELECT Balance FROM Account_Transactions WHERE Account_No = 'A123';

 balance
---------
     150

(1 rows)
cqlsh:banking>
```